Journal of Environmental Management 357 (2024) 120806



Contents lists available at ScienceDirect

Journal of Environmental Management

journal homepage: www.elsevier.com/locate/jenvman



Research article



Revolutionizing energy practices: Unleashing the power of artificial intelligence in corporate energy transition

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ARTICLE INFO

Keywords:
Artificial intelligence
Energy transition
Information asymmetry theory
Institutional theory
Corporate sustainable development

ABSTRACT

Corporate energy transition is crucial for long-term sustainable development. The widely discussed Artificial Intelligence (AI), as a disruptive technological innovation, is highly potential for enhancing environment performance. However, the specific impact of AI on the process of corporate energy transition and its underlying mechanisms have not been fully explored. This study focuses on A-share listed corporates in Shanghai and Shenzhen stock markets in China spanning from 2011 to 2021. Based on corporate annual report information and information from over 200,000 patent application texts, we innovatively construct indicators for corporate energy transition and AI technology application. Furthermore, we empirically investigate the impact of AI technology on corporate energy transition and its potential mechanisms through combining information asymmetry theory and institutional theory. The empirical results indicate that: 1) AI can drive corporate energy transition and the promoting effect of AI collaborative innovation on corporate energy transition should not be ignored. 2) AI can help corporates achieve energy transition through pathways such as mitigating information asymmetry, reducing financing constraints, adjusting sustainable development concepts and practices. 3) The driving effect of AI on corporate energy transition varies depending on the characteristics of different types of corporates, industries, and regions. This study provides strategic guidance and decision support for business managers and policymakers, assisting both corporates and governments in better utilizing AI technology during the social energy transition process to achieve a dual optimization of environmental and economic goals.

1. Introduction

Intergovernmental Panel on Climate Change (IPCC) *1.5 °C Special Report*[1] has explicitly stated that countries worldwide should achieve carbon neutrality by the mid-21st century. Among these, energy transition is a crucial aspect of upfront governance for carbon emission reduction. From the perspective of the United Nations Sustainable Goals (SDGs), energy transition not only contributes to reducing greenhouse gas emissions and mitigating climate change (SDG 13) but also ensures sustainable and modern energy access (SDG 7). Additionally, it aids in reducing energy poverty, enabling more people to access reliable and affordable energy supply (SDG1), thereby promoting inclusive growth in the socio-economic aspects of countries across the globe (Kouton, 2021). As the core practice of delivering the carbon peaking and carbon neutrality goals, energy transition has already garnered massive attention globally. For example, *The European Green Deal*[2] promulgated by the European Commission advocates the transformation of the overall energy system by greatly improving the use of renewable energy and energy efficiency, as well as the goal of zero net greenhouse gas emissions by 2050. Similarly, the *14*th *Five-Year Plan for Modern Energy System*[3] issued by the National Energy Administration of China. In fact, Corporate has now been the fastest growing energy demand for renewable resource (O'Shaughnessy et al., 2021), meanwhile, corporate is the primary entity of technological innovation. This implies that corporate

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[1] https://www.ipcc.ch/sr15/.
[2] https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en.
[3] https://www.gov.cn/zhengce/zhengceku/2022-03/23/content_5680759.htm.

https://doi.org/10.1016/j.jenvman.2024.120806
Received 11 February 2024; Received in revised form 31 March 2024; Accepted 31 March 2024
Available online 6 April 2024
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energy transition serves as a crucial lever to drive national energy transition and achieve dual carbon goals, playing a key role in promoting economic transformation, environmental protection, and sustainable development.

Corporate energy transition refers to the whole-process and whole-cycle transformation of energy acquisition, consumption and management through the introduction and adoption of environment-friendly technologies, policies and practices in the process of production, operation and even product sales (Sovacool, 2016). The energy transition of corporates can not only reduce the energy cost and improve their competitiveness, but also reduce the pollution to the external environment (Zhang and Kong, 2022). Through energy transition, corporates can reduce energy expenses and reduce risks caused by energy price fluctuations (Bolton et al., 2016). Most importantly, corporate energy transition can improve the social image of enterprises and attract more consumers and investors (Benlemlih et al., 2023), thereby creating opportunities for enterprises to obtain more economic investment. To sum up, corporate energy transition not only drives the green transformation of society, but also plays a crucial role in its own sustainable development. However, the existing literature mainly focuses on discussing energy transition at the city or even the more macro level (Lee et al., 2022b; Tzeremes et al., 2023), lacks in-depth exploration of corporate energy transition. This study explores the influencing factors and mechanisms of corporate energy transition, which can enrich the concerning literature from a micro perspective, helping deliver the Carbon Neutrality Goal and SDGs.

As the most representative technology in the Industry 4.0 era, Artificial Intelligence is a holistic transformative technology which has the potential to reshape the social and economic status of our society (Velarde, 2020). The abilities of Artificial Intelligence such as learning, decision-making and interacting with other entities surpass any other digital technologies (Internet, Information Communication etc.) (Zheng et al., 2017). The rise of Artificial Intelligence has already caused enormous impact on socio-economy and process of industrial production. Wang et al. (2022a) pointed out that the emergence of artificial intelligence is likely to enhance labor efficiency, but this may lead to a certain crowding-out effect in the labor market, resulting in a transformation of the types of employment in society. Meanwhile, they consider Artificial Intelligence is beneficial for industrialization and optimization of energy structure. The relevant application of AI like industrial robot may reduce the energy consumption by shortening the working hours, which eventually will advance energy productivity, according to many scholars (Chen et al., 2022). For corporates, the positive effect of AI on energy transition is based on practice. For example, the energy transition of a solar photovoltaic corporate in Netherland benefits from the application of AI (Schmidt et al., 2022); In Colombia, Hydroelectric power projects gradually achieved energy transition and reduced the economic risk through applying AI (Giraldo et al., 2021). However, in the spectrum of our investigation, only the technologies in Industry 3.0 are discussed from the prospective of technology in the concerning literature of energy transition, such as Internet (Li and Du, 2021). At present, no empirical research has been found to estimate the marginal impact of AI technology on the energy transition of corporates and the specific impact mechanism as a holistic reform.

If addressing the aforementioned real-world issues, accurately constructing two indicators: artificial intelligence technology and corporate energy transition, poses a challenge in this study. Numerous scholars have focused on the measurement methods for these indicators; however, there still exists a divergence of opinions. Mainstream methods for measuring corporate energy transition include energy efficiency (Li and Du, 2021), carbon emissions (Benlemlih et al., 2023), and energy structure (Lu et al., 2023), among others. However, these measurement methods start from the effects of energy transition to measure it. As mentioned earlier, energy transition, involving the entire process of corporate production, operation, and sales, should attempt to use a more comprehensive approach for measurement. However, existing literature has not made deeper or more innovative attempts at measuring corporate energy transition. Zhou and Li (2023), in exploring the relationship between corporate digital transformation and trade credit financing, used the frequency of occurrence of digital transformation-related keywords (identified through text analysis methods based on the Java PDFbox library) in corporate annual reports as a measurement indicator for digital transformation. This provides a new perspective for measuring corporate energy transition in this study. Regarding the portrayal of artificial intelligence, existing research characterizes the development of enterprise artificial intelligence mostly using industrial robots (Acemoglu and Restrepo, 2020). However, with the rapid iteration of industrial intelligence, relying solely on robot applications in the Industry 4.0 era may introduce certain biases in reflecting the development of enterprise artificial intelligence technology. According to Liu et al. (2022a), artificial intelligence algorithms include artificial neural networks, genetic algorithms, and particle swarm optimization algorithms, indicating that artificial intelligence encompasses various technologies comprehensively. Therefore, we reference the approach of Chu et al. (2024) and attempt to measure artificial intelligence using the more comprehensive method of patent quantity. For the method of identifying artificial intelligence patents, please refer to Section 3.2.2.

China is an excellent sample for exploring the impact of AI on corporate energy transition, this is mainly determined by two factors. Firstly, according to the World Energy Statistical Yearbook,[4] the total energy consumption in China will account for 26.4% of the global total in 2022, consistently holding the top position on the global list. Compared with post-industrialized countries such as the United States, Japan and United Kingdom, where energy consumption has begun to decline, China's achievement of energy transition seems to be a key factor in curbing global warming. To address this, China has proposed a national-level strategy to achieve carbon peaking by 2030 and the carbon neutrality by 2060. Secondly the Chinese government has recognized the crucial role of technological innovation, particularly artificial intelligence, in promoting energy transition and improving the ecological environment. Since the announcement of "Made in China (2015)" in 2015, which emphasized using AI to facilitate the upgrading of the manufacturing industry, several well-known Chinese enterprises, including the State Grid Corporation of China[5] and China National petroleum Corporation,[6] have significantly improved energy efficiency and reduced energy emissions through AI technology. Additionally, China filed a total of 445,000 AI patents from 2018 to 2021, ranking first globally, with enterprises being the main contributors to patent applications.[7] In summary, both Chinese government and corporates are actively seeking an effective path for achieving green and sustainable development across society through the disruptive technology of artificial intelligence.

Based on the above, we present the three research questions that this study aims to address: 1) Can artificial intelligence have an impact on corporate energy transition? 2) If so, what are the specific mechanisms through which artificial intelligence technology influences corporate energy transition? 3) Will the impact of artificial intelligence technology on corporate energy transition be subject to interference from other constraining factors? To answer these questions, this study combines data from Chinese A-share listed corporates from 2011 to 2021, utilizing machine learning methods to construct indicators for corporate energy transition and the application of artificial intelligence technology based on the textual information from annual reports and over 200,000 patent

[4] https://kpmg.com/cn/zh/home/campaigns/2023/10/statistical-review-of-world-energy-2023.html.

[5] http://paper.people.com.cn/zgnyb/html/2021-07/12/content_3057905.htm.

[6] http://center.cnpc.com.cn/bk/system/2023/07/18/030107107.shtml.

[7] https://caijing.chinadaily.com.cn/a/202112/16/WS61baad20a3107be4979fd52f.html.

application texts. The study systematically explores the impact and specific mechanisms of artificial intelligence on corporate energy transition. Our research holds significant implications for policymakers and business managers contemplating strategic arrangements to promote the sustainable development of enterprises through the empowerment of artificial intelligence.

The innovations of this study can be summarized in three main aspects: Firstly, our study explores the practical scenarios of corporate energy transition in the AI era and expands the applicability of institutional theory in the micro-level sustainable development of enterprises. Existing research on factors influencing corporate energy transition has deeply analyzed four dimensions: government policies (Zhang and Kong, 2022), market incentives (Lu et al., 2023), public pressure (Qiao and Yin, 2021), and technological progress (Du et al., 2023) (see Section 2.1.1 for a detailed analysis). However, artificial intelligence, as a technology variable with epochal characteristics and transformative capabilities, has not received sufficient attention in this field. To our best knowledge, this study is the first to explore the empirical research on the impact of artificial intelligence on corporate energy transition and its specific mechanisms. This provides practical references for enterprise managers in adopting sustainable development strategies. Simultaneously, we discuss the role of information asymmetry, financing constraints, sustainable development concepts, and practices in the impact of artificial intelligence on the path of corporate energy transition based on information asymmetry and institutional theory. Specifically, form the perspective of artificial intelligence, we enrich the implications of information asymmetry theory and institutional theory applied to the connotation of corporate energy transition. To sum up, information asymmetry theory and institutional theory manifest in entirely new forms during the process of enterprises achieving energy transition, innovating the mechanisms through which AI influences corporate energy transition. Therefore, this represents the theoretical and practical mechanism expansion of the aforementioned two theories in the field of AI's influence on corporate green transformation research.

Secondly, in terms of indicator measurement, this study starts from the concept of corporate energy transition and combining company annual reports with relevant official documents on energy transition at home and abroad, innovatively utilizes machine learning to construct indicators for measuring corporate energy transition. The aim is to capture the energy transition involved throughout the entire process of corporate operational development. On the other hand, this study combines the basic connotations of Industry 4.0 with machine learning methods, using the patent applications for artificial intelligence by listed companies in China to characterize the application of artificial intelligence technology. This method effectively enhances the measurement of micro-level corporate energy transition and the development of artificial intelligence, laying a solid foundation for subsequent related research.

Furthermore, in terms of research content and framework, this study enriches the literature on the impact of AI on social green transformation. We discuss the constraints (interference) factors of artificial intelligence in promoting corporate energy transition from various perspectives, including the corporate (ownership, corporate lifecycle, customer concentration, and environmental information disclosure), industry (degree of pollution), and region (environmental governance intensity). These analyses not only provide effective practical insights for government departments aiming to achieve societal green transformation at the industry and regional levels but also contribute to a deeper understanding within the academic community regarding the pathways through which artificial intelligence influences corporate energy transition. Given that many emerging economies are still in the early stages of exploring digital technology development and energy transition, heterogeneous experiences from Chinese enterprises will be beneficial for the practical application of AI technology in promoting societal green transformation in emerging economies.

2. Literature review and hypotheses development

2.1. Literature review

2.1.1. Influencing factors of corporate energy transition

As a crucial manifestation of corporate green transformation, energy transition involves various aspects of corporate behavior. This has led to the diversification of measurement methods for energy transition. For example, aspects such as the improvement of energy usage efficiency (Li and Du, 2021), reduction of carbon emissions (Benlemlih et al., 2023), innovation in energy technology (Du et al., 2023), and adjustment of energy structure (Lu et al., 2023) are considered.

Considering the above, research on factors influencing corporate energy transition can be broadly categorized into four types, namely government-driven, market-driven, public-driven, and technology-driven. It is important to note that discussions on relevant variables related to corporate energy transition in existing literature often take into account the aforementioned multiple factors. From the perspective of government-driven initiatives, existing literature can be classified into government regulation and government support. For instance, Zhang and Kong (2022), based on renewable energy policies, validate their long-term promoting impact on corporate energy transition, which, however, may be overshadowed by resource allocation efficiency and technological innovation. Additionally, government-driven research considers the impact of supply-side reforms, such as government subsidies, on the performance of corporate energy transition (Qi et al., 2022). They found that government subsidies can alleviate the pressure on R&D investment, thereby enhancing the performance of energy transition.

Research on market-driven corporate energy transition is also abundant, with a focus on the influence from financial markets. Taking the issuance of green bonds as an opportunity, (Benlemlih et al., 2023) explore the impact and mechanisms of market financing on corporate energy transition. They find that the issuance of green bonds significantly inhibits carbon emissions for companies, but this inhibitory effect has a one-year lag. Meanwhile, green finance policies, as a representative policy in financial markets, are considered an effective measure for environmental governance by various countries. Its inhibitory effect on carbon emission intensity in manufacturing companies has been extensively discussed by (Lee et al., 2022a). In addition, the role of digital inclusive finance in corporate energy transition should not be overlooked (Lu et al., 2023).

With the increasing awareness of environmental protection among the public and the gradual disclosure of environmental information by businesses and governments, decision-makers in enterprises are beginning to consider pressure from the public when formulating corporate strategies (Gu et al., 2021). This pressure can be divided into public demand and public concern. For example, Qiao and Yin (2021) found that consumers, as stakeholders in businesses, create a lucrative market for green products and services based on their demand, leading to a positive social reputation that encourages businesses to undergo energy transition. Tang and Tang (2016), drawing on agenda-setting theory and stakeholder theory, revealed that the media can significantly curb corporate pollution emissions by influencing public concern and government attention, indirectly demonstrating the importance of public pressure in corporate energy transition.

Compared to the first three driving forces, the influence of technology-driven factors in corporate energy transition has become a gradually emerging research branch in this field. This branch of research primarily focuses on the discussion of technology-driven energy transition based on digital technologies. For example, Du et al. (2023) used energy-related patents as a metric for measuring corporate energy innovation, exploring the impact mechanisms and constraints of digital transformation on energy innovation. They found that the positive effects of digital transformation on energy innovation vary due to differences in corporate ownership and the degree of energy technology

advancement. Li and Du (2021), based on the Internet Development Index, verified that the application of internet technology contributes to improving corporate energy efficiency, and they argued that leading companies in provinces exhibit significant technological spillover effects. In addition to digital technologies, Li et al. (2023) treated coal consumption as a crucial proxy variable for industrial enterprise energy transition, examining the impact of financial technology on it and its mechanisms (innovation performance and export performance). To better illustrate the research progress in relevant literature, we have created Table 1. It becomes apparent that the impact and mechanisms of AI on energy transition urgently need exploration.

2.1.2. The impact of artificial intelligence applications on enterprises

Existing research has indicated that artificial intelligence technology has been applied in various aspects of enterprise operations and management worldwide, including green supply chain management, reverse logistics, automation, and the Internet of Things (Chu et al., 2024; Dhamija and Bag, 2020). The impact of AI applications in enterprises can be broadly categorized into economic benefits and environmental sustainability. In terms of economic benefits, firstly, AI technology can enhance the efficiency of production, operations, and decision-making through intelligence, leading to increased revenue for enterprises (Gao, 2023). Moreover, AI facilitates changes in the production processes and service models of businesses, thereby improving satisfaction among consumers and other stakeholders (Grover et al., 2022). Furthermore, AI-based applications such as intelligent customer service and personalized recommendation systems leverage precise data acquisition and analysis to achieve more targeted market positioning (Huang and Rust, 2021). This provides businesses with opportunities to gain a deeper understanding of customer needs, better adapt to market changes, and create new business opportunities.

The impact of artificial intelligence technology on the environmental sustainability of businesses is also worth noting. Current research in this area primarily consists of two branches of literature. One is how artificial intelligence is smoothly implemented within enterprises to achieve sustainable development (Ghobakhloo et al., 2023). For instance, Ghobakhloo et al. (2023) discussed the feasibility of intelligent automation in enterprises from the perspectives of technology, organization, environment, and human resources. They find that the main factors influencing the implementation of intelligent automation are the internal environment of the enterprise, suggesting that companies with formal sustainable development strategies are more likely to transform intelligent automation into social, economic, and environmental performance. The second strand investigates the environmental consequences of the application of artificial intelligence in enterprises. For example, Benzidia et al. (2021) developed a conceptual framework and studied 168 hospitals in France, discovering a synergistic enhancement effect on environmental performance and green supply chain from the integration of big data analytics and artificial intelligence technology. Meanwhile, Liu et al. (2022b) measured the development level of artificial intelligence technology based on industry robot data and the number of academic papers related to artificial intelligence, finding that the impact of artificial intelligence on carbon intensity in the industrial sector is stage-specific and industry-heterogeneous. Furthermore, artificial intelligence not only addresses environmental pollution emissions from the source to the end but also has the potential to reduce health risks and urban security threats associated with pollution to some extent (Uddin et al., 2024). They point out that the reduction of these threats is mainly due to the alleviation of information asymmetry between enterprises and governments.

Based on the above review, several key insights have been identified. Firstly, existing literature predominantly focuses on outcome-oriented indicators such as emissions caused by energy, energy innovation, and energy consumption structure when assessing energy transition. Secondly, the role of artificial intelligence technology in the energy transition process at the enterprise level has not yet received sufficient and systematic discussion. In fact, research on energy transition primarily concentrates on the city level or even more macroscopic levels (Lee et al., 2022b; Tzeremes et al., 2023), with only a small fraction of literature delving into the influencing factors of corporate energy transition. These studies are limited to representative technologies of the Industry 3.0 phase, such as internet technology and information and communication technology, neglecting the comprehensive and disruptive significance of artificial intelligence technology.

Moreover, when exploring the impact of AI applications on enterprises, most research focuses on specific AI technologies or particular scenarios of AI technology applications, such as industrial robots (Chen et al., 2024), industrial intelligence (Wang et al., 2022b), etc. However, empirical evidence regarding the holistic impact of artificial intelligence technology on corporate energy transition is still lacking. Furthermore, the mechanisms through which artificial intelligence influences corporate energy transition remain unclear. Finally, the causal relationship between artificial intelligence and energy transition is yet to be explored, considering heterogeneous influences at the company level, industry level, and even the regional level.

2.2. Hypotheses development

As mentioned earlier, the impact of artificial intelligence technology on corporate energy transition has not yet received widespread attention from scholars. However, existing research still indirectly demonstrates the role of AI in corporate energy transition. From a macro perspective, artificial intelligence technology has already been widely applied in the energy systems of various countries and industries. Specifically, the manufacturing sector has extensively adopted industrial robot applications to enhance energy efficiency in relevant enterprises and industries, thereby promoting the transition of energy systems (Wang et al., 2022a). Some electronics companies also utilize digital platforms built with artificial intelligence technology for carbon footprint management, enabling the formulation of targeted carbon reduction strategies and facilitating the transition towards corporate energy-saving actions (Zhang et al., 2024). Simultaneously, a large energy conglomerate in

Table 1
Summary of literature on factors influencing corporate energy transition.

Authors	Year	Influencing factors: who is driving energy transition?				Measurement of energy transition
		Government	Market	Public	Technology	
Qiao and Yin	2020			×		Decision of energy transition
Li and Du	2021				Internet technology	Energy efficiency
Zhang and Kong	2022	×				Total factor productivity
Qi et al.	2022	×				Energy transition performance
Benlemlih	2022		×			Carbon emissions
Lee et al.	2022a		×			Carbon emission intensity
Lu et al.	2023		×			Energy structure
Du et al.	2023				Digital transformation	Energy innovation
Li et al.	2023				Fintech	Coal consumption

Source: compiled by authors

China has already utilized artificial intelligence technology to enhance its business divisions' capabilities in risk analysis, risk prediction, and traceability analysis (Ma and Chen, 2021). Undoubtedly, this contributes to the realization of corporate energy security transition.

Returning to the perspective at the enterprise level, although there is no direct empirical evidence indicating the impact of artificial intelligence on corporate energy transition, some practical cases indirectly suggest that AI may bring positive effects to corporate energy transition. Schmidt et al. (2022) conducted semi-structured interviews with small and medium-sized enterprises in the Netherlands engaged in solar photovoltaics and summarized the stages of their AI applications. They believe that most enterprises currently fall into the first stage of AI application and affirm the positive role of AI in energy transition. Furthermore, some scholars even assert that artificial intelligence is the key driving force for enterprises to achieve continuous operational transformation (Maslak et al., 2021).

Considering the energy transition defined in this study involves various stages from production to sales in the entire lifecycle of enterprises, we also attempt to demonstrate the positive effects of AI on corporate energy transition from three aspects: production, supply chain management, and product sales. Firstly, the application of AI in production equipment can achieve intelligent adjustment of devices through smart control systems, and this kind of transition of energy systems could directly enhance energy utilization efficiency (Ahmad et al., 2022). Secondly, enterprises can use AI algorithms to optimize logistics routes, improve transportation efficiency, and reduce energy consumption and carbon emissions (Tsolakis et al., 2022), thus advancing the refinement of the energy-saving actions transition system. Additionally, through big data analysis and machine learning, enterprises can more accurately predict market demand, optimize production plans, reduce excess capacity, and minimize energy consumption (Yue et al., 2021). This capability for big data prediction is beneficial for coordinating corporate energy security transition.

In fact, from the theoretical perspective, for those enterprises are striving to achieve sustainable development through energy transition, both economic resources and the concept of sustainable development are indispensable. Based on the information asymmetry theory and institutional theory, the acquisition of economic resources and the adjustment of sustainable development strategies are influenced by information asymmetry issues and institutional pressures from stakeholders. The role of AI technology or digital technology in addressing information asymmetry issues and alleviating institutional pressures has received some initial attention from scholars (Horvitz et al., 1988; Zhang et al., 2024). Hence, we propose first hypothesis.

H1. Artificial intelligence technology has significant positive impact on corporate energy transition.

According to the theory of information asymmetry, information asymmetry among stakeholders in the market may lead to financing problems (Mishkin, 1990). Existing research suggests that digital technology provides a unique opportunity to reduce information asymmetry between companies and the market. On one hand, the digital transformation of companies and the application of digital technology can promote effective integration of information (Vassileva et al., 2012) and enhance information flow among companies (Iqbal et al., 2018). On the other hand, the emergence of various technology-driven information platforms, including artificial intelligence technology, encourages companies to disclose information to external stakeholders, further boosting the confidence of external investors (Fenwick et al., 2019). However, the above studies have not empirically examined the role of different technologies in reducing information asymmetry from the perspective of empirical experience. A small number of scholars have verified the significant reduction in information asymmetry through big data analytics technology based on data from Jordan (Zhang et al., 2024). It is worth mentioning that Cui et al. (2022) demonstrated from the perspective of intangible assets of artificial intelligence that the application of artificial intelligence can improve corporate governance by reducing the degree of information asymmetry, providing strong theoretical support for the mechanism hypothesis of this study. Considering the focus of this study on the development degree of artificial intelligence technology, we still believe that it is essential to reconfirm the relationship between artificial intelligence technology and information asymmetry.

If the issue of information asymmetry between enterprises and stakeholders such as financial institutions can be effectively alleviated, companies may have the opportunity to obtain more economic resources. This can be elaborated from the perspectives of investors and financial institutions. From the investor's perspective, enhancing the quality and frequency of information disclosure through public financial statements, operational reports, and future plans can reduce information asymmetry, thereby increasing investor confidence and willingness to invest (Diantimala et al., 2022).

From the perspective of financial institutions, enterprises can leverage blockchain technology to communicate and negotiate credit application information with financial institutions in a timely manner (Javaid et al., 2022). Moreover, when banks and other financial institutions can access higher-quality "soft" and "hard" information from enterprises, they can make credit decisions more swiftly, directly reducing the overall financing costs for enterprises. For corporates, with the premise of maintaining stable development, an increase in economic resources will encourage enterprises to purchase more green equipment and participate in energy transition practices (Tan et al., 2024). Furthermore, during the loan renewal process between banks and enterprises, banks and other financial institutions can utilize their vast proprietary information, and the stringent requirements of green credit will incentivize enterprises to engage in green practices (Houston and Shan, 2022). Therefore, we propose the following hypothesis.

H2. Artificial intelligence technology can alleviate financing constraints by reducing information asymmetry, thereby promoting corporate energy transition.

Institutional theory suggests that the survival of businesses is not isolated, and their operations and strategic decisions are influenced by the external environment (Fernando and Lawrence, 2014). In real life, the criteria for balancing interests between external and internal stakeholders and business managers often diverge. External institutional pressures, such as "green" pressure from stakeholders, are a typical form of external institutional pressure. The digital transformation and technological innovation of businesses have a dual effect on the institutional pressures they face during the energy transition process. From a production and operational perspective, the use of technologies like artificial intelligence and robotics may replace a significant number of low-skilled workers (Webb, 2019). This trend promotes standardization in the production process and transparency in emission information. As the manipulative space of businesses becomes smaller, and external regulatory costs decrease, stakeholders are likely to exert more "green" pressure on business managers. Managers may, to some extent, passively adjust sustainable development strategies to meet some of the stakeholders' demands (Siebenhüner and Arnold, 2007). From an efficiency perspective, new technologies, including AI, make investment behavior more efficient and decision-making processes more precise (Horvitz et al., 1988). Businesses may become increasingly willing to engage in innovative practices, enhancing performance while expanding their scale (Farboodi et al., 2019). In this scenario, business managers, facing "green" pressure from stakeholders, may be more inclined to proactively adjust their sustainable development strategies to meet their demands.

The adjustment of corporate sustainable strategies is often reflected in two aspects: ideology and practice. At the ideological level, climate risk perception and a company's ESG performance are often used to measure the company's concept of sustainable development. Existing literature has demonstrated that climate risk perception (Kouloukoui

et al., 2021) will prompt companies to reconsider and evaluate their long-term strategies, possibly making climate change a core business issue. At the same time, companies with stronger ESG performance are likely to be more inclined towards energy transition, a viewpoint acknowledged by scholars (Wang et al., 2024). On the practical level, indicators such as green investments, energy consumption, and pollution emissions (Haque and Ntim, 2018) are often seen as crucial metrics for sustainable development practices. Chen et al. (2023) pointed out that an increase in corporate green investments stimulates increased green innovation activities, thereby encouraging companies to actively engage in energy transition. Based on the above, we propose the final hypothesis.

H3. The Artificial Intelligence technology can accelerate corporate energy transition by adjusting both the ideology and practices of sustainable development.

Based on the contents of Sections 1 and 2, we construct a research framework, as illustrated in Fig. 1.

3. Methodology

3.1. Model Specification

This study primarily examines the impact of artificial intelligence on corporate energy transition, establishing the following model:

$$Ln_ET_{it} = \alpha + \beta Ln_AI_{it-1} + \gamma X_{t-1} + \gamma_i + \theta_t + \mu_{it} \quad (1)$$

Where Ln_ET_{it} represents corporate energy transition i in year i, Ln_AI_{it} denotes the artificial intelligence technology innovation of corporate i in year $i-1$. X_{t-1} denotes a set of control variables at the corporate level, γ_i and θ_t represents the corporate and year fixed effects; μ_{it} is the random error term.

3.2. Variable

3.2.1. Dependent variable: Corporate energy transition

In China, national policies and strategic practices play a crucial role in guiding and influencing domestic enterprises. Information related to energy transition is often reflected in the strategic corporate annual reports and corporate social responsibility reports. Terms such as "energy conservation" and "consumption reduction" frequently signify a company's strategic initiatives and achievements in energy transition. In recent years, there has been a significant increase in literature utilizing textual data for empirical analysis to extract key economic information. It is noteworthy that the studies by, Hassan et al. (2019) and Florackis et al. (2023) successfully measured company-level political and cybersecurity risk indicators using information from annual reports. Therefore, we posit that a method based on the frequency analysis of terms in corporate annual reports can effectively gauge the extent of energy transition within enterprises.

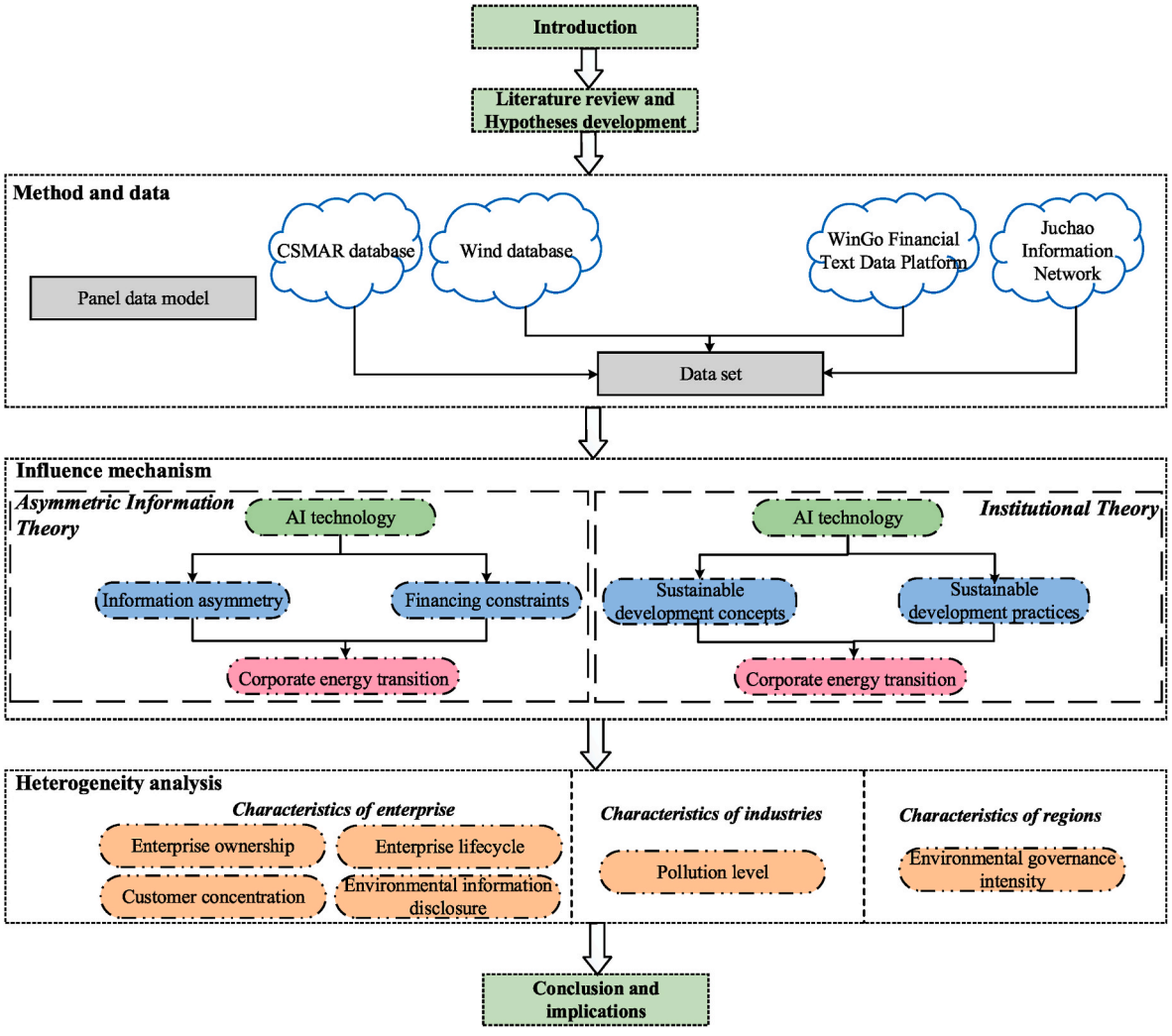


Fig. 1. Research framework.

Firstly, from the perspective of the entire process of corporate energy transition, it is an ongoing initiative for companies to implement climate transition strategies and engage in energy-saving practices. We utilize the *JavaPDFBox* library to extract key indicators from environmental information reports and Chinese environment reports published by international organizations such as the OECD and the World Bank. In order to ensure the relevance and scientific validity of these seed terms to our research, further keyword analysis is conducted using the Delphi method to establish consensus. Validation is obtained through expert identification, resulting in a total of 68 seed keywords associated with energy transition.

Next, these keywords are input into the vector layer through Python, utilizing the Continuous Bag of Words (CBOW) model, an extension of the *Word2Vec* neural network algorithm in machine learning, to expand the seed word corpus by incorporating similar words. We focus primarily on the Management Discussion and Analysis (MD&A) section in corporate annual reports. The CBOW model is specifically represented as:

$$max\sum_{k\in Z}\log p(k|Context(k)) \tag{2}$$

Where k represents the central word, Z represents the corpus, and Context(k) represents the context before and after the central word. After processing with the Word2Vec algorithm, an expanded vocabulary is obtained, ultimately identifying 285 keywords related to corporate energy transition. Through the machine learning LDA clustering analysis process, these keywords are classified into three dimensions, including corporate energy system transition, corporate energy-saving actions transition, and corporate energy security transition, with 92, 105, and 88 keywords respectively. Furthermore, to address the disclosure lag issue in corporate annual reports, this study employs a rolling calculation method to calculate the keyword frequencies of specific texts in annual reports for a particular year and its one-year before and after periods, taking the arithmetic average. Finally, this research used the natural logarithm transformation plus one to avoid zero counts and measured the energy transition level of listed companies as the dependent variable.

3.2.2. Independent variable: artificial intelligence technology

Corporate Artificial Intelligence Technology Innovation (*Ln_AI*) is measured by the number of patents related to artificial intelligence that a company applies for in the current year. Many studies have used industry data on industrial robots as an indirect measure of the level of enterprise artificial intelligence application (Acemoglu and Restrepo, 2020). However, utilizing indirect identification strategies with industry data may potentially introduce biases.

In view of this, this study proposes a new measurement method, outlined as follows: first, combining the International Patent Classification (IPC) numbers from the 'Digital Economy Core Industry Classification and International Patent Classification Reference table (2023) (https://www.gov.cn/zhengce/zhengceku/2023-04/11/content_5750914.htm)' (referred to as the Reference Table), we use Python software to match the IPC classification numbers of patents applied by listed companies. Second, we conduct text analysis on 15 domestic and international documents and reports related to artificial intelligence development. After meticulous manual filtering, we obtain 73 seed words. Subsequently, based on the text corpus of patents applied by listed companies, we use the Word Embedding machine learning method to expand the seed words. After further manual refinement, we obtain 252 keywords related to artificial intelligence. Using these keywords, we identify and match artificial intelligence patents in corporate patent texts. By combining these two methods, which complement each other, we comprehensively consider the relevant patents of artificial intelligence technology in listed companies. We measure the development of artificial intelligence technology in companies by the number of artificial intelligence technology patents applied for in the current year (processed with a natural logarithm, plus one).

3.2.3. Control variable

Referring to Du et al. (2023) and Zhao et al. (2022), this study controls for the following variables that may influence corporate energy transition: specifically, firm size (*Size*), leverage (*Lev*), return on assets (*Roa*), firm age (*Age*), growth (*Growth*), capital intensity (*Fix*), equity structure (*Top1*), board independence (*Indep*), dual leadership structure (*Dual*), and board size (*Board*). For variable descriptions, please refer to Table 2.

3.3. Data source and descriptive statistics

Our research sample consists of A-share listed companies on the Shanghai and Shenzhen stock exchanges in China from 2011 to 2021. The textual data for corporate patents is sourced from the WinGo Financial Text Data Platform, while the annual report data is sourced from the Juchao Information Network. Other company characteristic data is mainly obtained from the CSMAR and Wind databases. The following data processing steps are undertaken: 1)To mitigate the impact of extreme values, a two-tailed trimming is applied to continuous variables at the 1% level. 2)Exclusion of the primary research data with severe missing values, companies listed for less than one year, and samples from industries such as finance, insurance, and real estate. 3) Removal of companies with abnormal operational conditions, such as ST (Special Treatment), PT (Potential Special Treatment), and ST* listings. These procedures resulted in a final dataset comprising 24,870 observations at the company-year level.

The descriptive statistics for the main variables in this study are presented in Table 3. The mean value of the dependent variable, corporate energy transition performance, is 1.9833, with a standard deviation of 0.9477. The minimum value is 0, and the maximum value is 5.6525, indicating a relatively balanced but overall improvable level of energy transition development among Chinese listed companies. The explanatory variable, artificial intelligence technology (*Ln_AI*), has a mean value of 0.0965, with a standard deviation of 0.3779. The

Table 2
Symbols and definitions of variables.

Variable	Name	Symbol	Definition
Dependent Variable	Corporate Energy Transition	*Ln_ET*	ln (1+Number of Energy-related Terms in MD&A Section)
Independent Variable	Artificial Intelligence Innovation	*Ln_AI*	ln (1+Number of Artificial Intelligence Patents)
Control Variable	Corporate Scale	*Ln_Size*	ln (Logarithm of Total Assets (in CNY))
	Return on Assets	*ROA*	Net Profit (in CNY)/Average Total Assets (in CNY)
	Debt-to-Equity Ratio	*LEV*	Total Liabilities (in CNY)/Total Assets (in CNY)
	Years Listed	*Ln_Age*	ln (Current Year - Year of Listing + 1)
	Revenue Growth Rate	*Growth*	(Current Operating Income (in CNY) - Previous Operating Income (in CNY))/Previous Operating Income (in CNY)
	Equity Concentration	*Top1*	Ownership Percentage of the Largest Shareholder
	Per Capita Net Fixed Assets	*Ln_ Fix*	ln ((Net Fixed Assets (in CNY)/ Total Number of Employees)
	Dual Leadership Structure	*Dual*	If Chairman and CEO are the Same Person, Assign 1, Otherwise 0
	Board Size	*Ln_Board*	ln (Board Size)
	Board Structure	*Indep*	Ratio of Independent Directors to Total Board Size

Table 3
Descriptive statistics.

Variables	Obs	Mean	Std. Dev.	Min	Max	VIF
Ln_ET	24,870	1.9833	0.9477	0	5.6525	/
Ln_AI	24,870	0.0965	0.3779	0	2.3979	1.08
Ln_Size	24,870	22.2443	1.2803	19.5050	26.3951	1.86
ROA	24,870	0.0380	0.0653	−0.4147	0.2442	1.31
LEV	24,870	0.4406	0.2043	0.0274	0.9901	1.64
Ln_Age	24,870	2.1981	0.7546	0.6931	3.4340	1.33
Growth	24,870	0.1739	0.4474	−0.7316	4.8058	1.08
Ln_Fix	24,870	12.5621	1.1544	4.1274	19.5426	1.12
Top1	24,870	0.3451	0.1482	0.0838	0.7584	1.09
Dual	24,870	0.2641	0.4409	0	1	1.10
Ln_Board	24,870	2.1338	0.1994	1.6094	2.7081	1.56
Indep	24,870	0.3751	0.0535	0.3000	0.6000	1.43

minimum value is 0, and the maximum value is 2.3979, suggesting an overall low level of artificial intelligence technology development among listed companies, with significant differences in AI development between companies. The dispersion and differentiation levels of other control variables are generally consistent with existing literature.

4. Results of empirical analysis

Building upon the formulated hypotheses and methodology, this section comprises benchmark regression results, addressing endogeneity issues, analyzing the mechanisms through which artificial intelligence innovation affects both companies and energy transition, conducting heterogeneity analysis, and presenting further analysis.

4.1. Results of baseline regression

Table 4 presents the results of the baseline regression. The table consists of four columns, where columns (1) and (2) utilize the panel OLS method for estimation, with column (2) including all control variables in addition to those in column (1). Columns (3) and (4), in comparison to the preceding two columns, incorporate year and firm fixed effects.

Table 4
Results of baseline regression.

Variables	(1)	(2)	(3)	(4)
	Ln_ET	Ln_ET	Ln_ET	Ln_ET
Ln_AI	0.2548***	0.0620**	0.0756***	0.0594***
	(0.0320)	(0.0302)	(0.0183)	(0.0174)
Ln_Size		0.2454***		0.2140***
		(0.0130)		(0.0182)
ROA		−0.4811***		0.3093***
		(0.1380)		(0.0827)
LEV		−0.0372		−0.1073*
		(0.0703)		(0.0601)
Ln_Age		−0.0847***		−0.1169***
		(0.0172)		(0.0291)
Growth		0.0226*		0.0342***
		(0.0130)		(0.0078)
Ln_Fix		0.1676***		0.0630***
		(0.0122)		(0.0117)
Top1		−0.4649***		0.1809*
		(0.0853)		(0.1049)
Ln_Board		−0.3226***		0.0770
		(0.0736)		(0.0662)
Indep		−1.0028***		0.0713
		(0.2504)		(0.1787)
Dual		0.0504**		−0.0088
		(0.0250)		(0.0164)
Constant	1.9587***	−4.1594***	1.9760***	−3.5381***
	(0.0141)	(0.3184)	(0.0018)	(0.4389)
Observations	24,870	24,870	24,870	24,870
Adjusted R^2	0.0103	0.1583	0.7792	0.7905
Year FE	No	No	Yes	Yes
Firm FE	No	No	Yes	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

As per the results in Table 4, columns (1)–(4), the coefficient of *Ln_AI* remains consistently positive and statistically significant at least at the 10% level. Taking the example of column (4), the coefficient of *Ln_AI* is 0.0594 and is statistically significant at the 1% level. Specifically, a 1% increase in AI corresponds to a 0.06% increase in ET. This implies that strengthening innovation in artificial intelligence technology accelerates corporate energy transition. Thus, ***H1*** is confirmed.

Moreover, according to column (4), we observe that the majority of control variables pass significance tests and exert a significant impact on corporate energy transition, with the exception of *Dual* (Two Positions in One), *Ln_Board* (Board Size in Natural Logarithm), and *Indep* (Board Independence). Specifically, an increase in firm size, return on assets, revenue growth rate, and per capita net fixed assets, as well as a tendency towards concentrated equity, all contribute to the promotion of corporate energy transition. This finding aligns with Du et al. (2023). On the contrary, companies with longer listing durations and higher debt-to-equity ratios find it more challenging to undertake energy transition initiatives (Gao et al., 2022; Qiao et al., 2022).

Information sharing is recognized as a crucial pathway for enhancing the quality of collaborative research and development in enterprises. Numerous theoretical studies consider the degree of information sharing as a prerequisite for comparing the quality of collaborative and independent research and development outcomes. High levels of information sharing lead to greater innovation quality in collaborative research and development compared to independent research and development (Kamien et al., 1992). In collaborative research and development, companies share information, knowledge, and technology with their partners, not only enhancing the efficiency of knowledge acquisition but also fostering inspiration and innovation quality in the collaborative process (De Faria et al., 2010).Therefore, we further investigate whether collaborative research and development in the field of artificial intelligence also influence a company's energy transition. Specifically, we categorize collaborative AI innovation into collaborative AI invention patents and collaborative AI utility patents. According to the results in columns (1) and (2) of Table 5, both invention and utility patents in collaborative AI innovation contribute to promoting corporate energy transition. This indicates that engaging in collaborative research and development with other institutions can lower financing costs for companies and facilitate the integration of various innovative ideas during the R&D process (De Fuentes and Dutrénit, 2012). Consequently, this enhances the quality of AI innovation, driving corporate energy transition.

Table 5
Results of further analysis.

Variables	(1)	(2)
	Collaborative AI invention patents	Collaborative AI utility patents
	Ln_ET	*Ln_ET*
Ln_AI	0.0200**	0.0380***
	(0.0097)	(0.0075)
Constant	−3.4975***	−3.3829***
	(0.4373)	(0.4341)
Observations	24,870	24,870
Adjusted R^2	0.7903	0.7908
Controls	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

4.2. Endogenous test

While the preceding sections have provided preliminary evidence that the application of artificial intelligence technology contributes to promoting corporate energy transition, this conclusion may face challenges such as reverse causality and omitted variables, raising concerns about endogeneity. On one hand, if a company exhibits proactive performance in energy transition, the demand for artificial intelligence technology driven by energy consumption control requirements during the production process may be higher. Thus, it could potentially stimulate innovation in artificial intelligence. To address this issue, we employ instrumental variable methods. On the other hand, apart from the control variables considered in this study, energy transition in companies could be influenced by various internal and external factors, such as environmental regulatory policies and regional economic development levels. To mitigate potential omitted variable bias, we employ Propensity Score Matching with Difference-in-Differences (PSM-DID) and placebo tests. The specific tests are as follows.

4.2.1. Instrumental variables method (IV)

Firstly, drawing inspiration from Acemoglu and Restrepo (2020), we adopt the U.S. industry-level stock of industrial robots as a proxy for robot application in the Chinese industry. We construct an interaction term (*iv1*) by taking the product of the stock of U.S. industrial robots and the mean of AI innovation in other Chinese firms within the same industry and year. The results of the regression are presented in columns (1) and (2) of Table 6.

Secondly, recognizing sensors as critical components of robots and fiber optics as a crucial material for manufacturing sensors, there exists a close association between the density of fiber optics and AI innovation. Simultaneously, the deployment of fiber optics is considered exogenous and unaffected by corporate energy transition. Hence, we employ the length of long-distance fiber optics per 10,000 square kilometers (*iv2*) as another instrumental variable. The regression results are shown in columns (3) and (4) of Table 6.

4.2.2. PSM-DID

Based on the median of AI-related patents, we categorize enterprises into two groups: high AI innovation and low AI innovation. Subsequently, employing the control variables from previous sections as covariates, we conduct a 1:1 matching. Balance tests indicate that the majority of covariates exhibit imbalances of less than 5%, and t-tests reveal no significant differences between the treatment and control groups at the 10% significance level. This suggests good matching effectiveness for the sample data. Subsequent regression tests are then conducted on the matched sample, with the results presented in column (5) of Table 6.

4.2.3. Placebo test

This study draws inspiration from Cantoni et al. (2017) and employs an indirect placebo test to assess the impact of unobservable factors on the regression results. Specifically, the variable measuring the application of artificial intelligence technology in companies (*Ln_AI*) is randomly shuffled to create a new set of pseudo-treatment data. Regression is then performed 500 times on the randomly generated data, and the coefficients of the resulting error terms are estimated. The distribution of these coefficients is plotted for observation (Fig. 2). By observing the distribution plot, it is evident that the coefficients are clustered around zero, exhibiting significant differences from the baseline regression results. Moreover, the p-values are predominantly greater than 0.1. This indicates that the baseline regression results are robust and not significantly influenced by other unobservable random factors, providing robustness to the conclusions drawn in this study.

4.3. Robustness test

4.3.1. Alternative variables

For the core explanatory variable, we utilize two measurement approaches. Firstly, we employ a machine learning method to reconstruct a

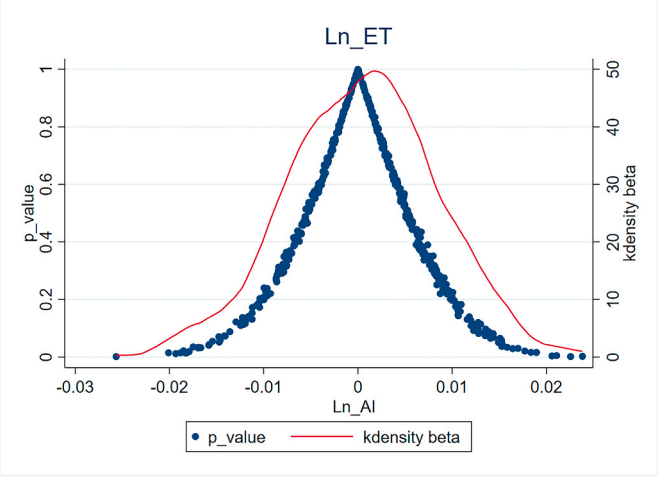


Fig. 2. Placebo test result of *Ln_ET*.

Table 6
Results of robustness tests: instrumental variable method and PSM-DID test.

Variables	(1)	(2)	(3)	(4)	(5)
	Ln_ET	Ln_ET	Ln_ET	Ln_ET	Ln_ET
Ln_AI		0.8597**		1.5009***	0.0619**
		(0.3616)		(0.5546)	(0.0255)
iv1	0.0017***				
	(0.0004)				
iv2			0.0048***		
			(0.0013)		
Constant	–	–	–	–	−3.1811***
					(0.9729)
K–P rk LM statistic	17.05***		13.82***		
C-D Wald F statistic	60.54		61.88		
K–P Wald rk F statistic	17.63		14.22		
Observations	24,671	24,671	24,671	24,671	6278
Adjusted R^2	–	–	–	–	0.7813
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

keyword library containing 336 terms from 30 policy documents related to energy transition both domestically and internationally. This library is then used to re-identify AI innovations in patents, resulting in a new AI innovation indicator (*Ln_AI_2*). Secondly, we replace the original independent variable with a dummy variable indicating whether a company possesses an AI patent (*AI_dummy*). The results of these two approaches are presented in columns (1) and (2) of Table 7. Regarding the dependent variable, we also expand the keyword library related to 'energy transition' and re-measured the dependent variable accordingly. The specific regression results are presented in column (3) of Table 7.

4.3.2. Alternative models

Considering *Ln_ET* is a count variable, this study employs a high-dimensional fixed-effects panel Poisson model, which is more suitable for skewed dependent variables, for analysis. Detailed results can be found in column (4) of Table 7. Furthermore, external influences such as macro systemic risks and industry cycle changes may impact corporate strategic decisions. Hence, we further control for 'province × year' and 'industry × year' fixed effects to address potential omitted variable issues. The regression results considering cross-fixed effects are presented in columns (5) and (6) of Table 7.

4.4. Mechanism analysis

After validating the significant and robust positive correlation between AI technology and corporate energy transition, we attempt to verify the causal relationship between them through mechanism analysis. Specifically, in this section, we examine potential mechanisms based on ***H2*** and ***H3*** proposed according to stakeholder theory, information asymmetry theory, and institutional theory. We investigate mechanisms such as information asymmetry, financing constraints, bank loans, climate risk perception, sustainable development principles, and green investments.

4.4.1. Improving information asymmetry and alleviating financing constraints

Based on stakeholder theory and information asymmetry theory, a company's sustainable development behavior is influenced by various external entities, including external financial institutions. Regarding the implementation of sustainable development strategies through corporate financing, the financing behavior of external financial institutions is influenced by factors such as information asymmetry. This information asymmetry can be mitigated to some extent through the development of information technology (Cheng et al., 2021). Hence, considering the backdrop of AI technology development, we propose ***H2a*** and ***H2b***. To test these hypotheses, following the approach of Teng and Tan (2023) and Chen and Dagestani (2023), we regress *Ln_AI* on four intermediary variables (*ASY1*, *ASY2*, *SA*, and *Ln_Load*). Refer to Table 8 for detailed results.

Firstly, AI technology has the potential to mitigate information asymmetry to a certain extent, thereby promoting corporate energy transition. To validate this hypothesis, we employ two calculation methods to measure the level of information asymmetry. Firstly, following the approach of Amihud et al. (1997), we use daily trading data to calculate the illiquidity ratio as an indicator of information asymmetry (*ASY1*). The formula is: $ASY1_{it} = -\frac{1}{D_{it}}\sum_{k=1}^{D_{it}}\sqrt{\frac{|r_{it}(k)|}{V_{it}(k)}}$. Secondly, referring to Bharath et al. (2009), we extract the first principal component of illiquidity ratio (*ASY1*), liquidity ratio (*LR*), and earnings reversal indicator (*GAM*) as a proxy variable for information asymmetry (*ASY2*). Here, $LR_{it} = -\frac{1}{D_{it}}\sum_{k=1}^{D_{it}}\sqrt{\frac{V_{it}(k)}{|r_{it}(k)|}}$, where $r_{it}(k)$) represents the stock return on the k-th trading day of company i in year t, $V_{it}(k)$ represents the daily trading volume, and D_{it} is the number of trading days in the year. The larger the values of the two information asymmetry indicators (*ASY1* and *ASY2*), the more severe the information asymmetry. According to columns (1) and (2) of Table 8, the coefficients of *Ln_AI* on *ASY1* and *ASY2* are −0.0241 and −0.0465, respectively, both passing the 1% significance test. This implies that the increased application of AI technology can alleviate information asymmetry between companies and external entities. The establishment of multiple technological platforms empowered by AI technology enhances information exchange between companies and the public, financial institutions, and other social entities. Based on this, the reduction of information asymmetry between companies and other social entities will lead to two scenarios:

Table 8
Results of mechanism analysis: the role of information asymmetry, financing constraints, and bank loans.

Variables	(1)	(2)	(3)	(4)
	ASY1	ASY2	SA	WW
Ln_AI	−0.0241***	−0.0465***	−0.0157***	−0.0175*
	(0.0089)	(0.0162)	(0.0029)	(0.0097)
Constant	3.0388***	4.8327***	3.6666***	−0.1734
	(0.1418)	(0.2430)	(0.1097)	(0.2402)
Observations	24,870	24,870	24,870	24,870
Adjusted R^2	0.7391	0.7218	0.9608	0.8640
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

Table 7
Results of robustness tests: remeasuring variables and replacing models.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	Ln_ET	*Ln_ET*	*Ln_ET_2*	*Ln_ET*	*Ln_ET*	*Ln_ET*
Ln_AI			0.0591***	0.0165**	0.0712***	0.0729***
			(0.0226)	(0.0081)	(0.0174)	(0.0173)
Ln_AI_2	0.0333***					
	(0.0123)					
AI_dummy		0.0612***				
		(0.0180)				
Constant	−3.5215***	−3.5479***	−2.6987***	−1.9149***	−3.6609***	−3.5043***
	(0.4375)	(0.4385)	(0.5681)	(0.2392)	(0.4121)	(0.4158)
Observations	24,870	24,870	24,870	24,862	24,870	24,870
Adjusted R^2	0.7904	0.7904	0.7457	0.1224	0.8065	0.8081
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind × Year FE	No	No	No	No	Yes	Yes
Pro × Year FE	No	No	No	No	No	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

companies with good environmental performance will receive more economic resources to support their sustainable development (such as energy transition) (Bassetti et al., 2021); companies with relatively poor environmental performance will face public pressure from external sources, forcing them to adjust their sustainable development strategies and promote energy transition (Abreu et al., 2015).

Secondly, AI technology can alleviate financial constraints to a certain extent, and thereby promoting their energy transition. Similarly, we use the SA index (*SA*) (Hadlock and Pierce, 2010) and WW index (*WW*) as two variables to measure financial constraints. From columns (3) and (4) of Table 8, we observe that AI technology significantly reduces corporate financial constraints. The reason for this phenomenon may be that the application of AI technology not only enhances the transparency of corporate information but also, to some extent, wins high expectations of future earnings from financial institutions by improving its own productivity. According to Tan et al. (2024), companies with more economic resources are likely to allocate more funds to sustainable development practices. Based on the above, ***H2*** is confirmed. From practical experience, scholars have found that artificial intelligence-intensive companies, tend to have stronger capabilities in terms of capital acquisition and profitability compared to other types of companies (Wamba-Taguimdje et al., 2020). Therefore, they have greater economic potential to implement new environmental remediation projects (Moro-Visconti et al., 2023). Additionally, Koley (2021) found that Indian manufacturing companies during the pandemic era also tend to adopt advanced technologies, including artificial intelligence and big data, to alleviate the shortage of human capital. The cost savings from these measures are more likely to be invested in the construction of green-related projects, thus promoting the sustainable development of enterprises (Moro-Visconti et al., 2023). This suggests that the conclusions of subsequent mechanism and heterogeneity analyses in this study may not only apply to enterprises in China but also provide policy insights for other emerging countries that urgently need to improve environmental quality and achieve sustainable development goals through artificial intelligence technology.

4.4.2. *Adjusting sustainable development concepts and practices*

According to stakeholder theory and institutional theory, while facing issues such as information asymmetry, companies may receive additional advice and requests from stakeholders regarding strategic adjustments, which can influence the strategic decisions of company managers. Based on this, this section attempts to discuss: in the context of external institutional pressures, can AI technology affect a company's strategic philosophy and decision-making, and whether such strategic adjustments will promote the company's energy transition. To test ***H3a*** and ***H3b***, we select relevant mediating variables from the aspects of sustainable development concepts and practices. The regression results are detailed in Table 9.

Firstly, AI technology can prompt companies to adjust their sustainable development philosophy, thereby accelerating energy transition. To validate the role of sustainable development philosophy in the impact of AI technology on energy transition, we adopt research approach of Li et al., 2020. Combining machine learning methods to analyze the MD&A section of company annual reports, we measure the company's perception of climate risk (measured by the proportion of relevant keywords in the MD&A section). Simultaneously, this study uses the Shanghai Huazheng ESG rating index to measure a company's sustainable development philosophy. The regression results for climate risk perception are shown in Table 9, column (1), indicating that the coefficient of *Ln_AI* on Climate is 0.0097 and passes the 1% significance test. This suggests that AI technology application can enhance managers' perception of climate risk, and managers with high climate risk perception tend to engage in green transformation practices (Kouloukoui et al., 2021). For the ESG variable, we disaggregate the sustainable development philosophy into three dimensions: environmental, social, and governance, and estimated coefficients for each. The relevant results are shown in Table 9, columns (2) to (5). It is evident that a company's sustainable development philosophy (ESG) can indeed be significantly promoted by the application of AI technology. However, compared to the environmental (*E_score*) and governance (*G_score*) dimensions, the social responsibility philosophy (*S_score*) does not seem to be effectively influenced by AI technology. This indicates that AI technology application can facilitate stakeholders to exert more 'green' pressure, prompting companies to promptly adjust their sustainable development philosophy. This sustainable development philosophy is mainly reflected in the environmental and corporate governance aspects. As a crucial manifestation of corporate sustainable development, the role of energy transition cannot be ignored, and companies with a strong sustainable development philosophy are more likely to engage in green transformation activities, including energy transition (Wang et al., 2024).

Next, AI technology can accelerate the energy transition process by adjusting sustainable development practices. This study comprehensively measures sustainable development practices using indicators such as green investment, energy consumption, the number of pollution types, and the number of pollution energy consumption types. Green investment is constructed by adding 1 to the total environmental investment in management expenses and the total environmental investment in ongoing projects, and then taking the logarithm. Following Lee et al. (2022a), this study calculates the energy consumption intensity of enterprises using the formula $ECI_{it} = \frac{encn_{it}}{toput_{it}}$, Where *encn* refers to the enterprise energy consumption, and *toput* refers to the total operating income of the enterprise. The number of pollution types (*PT*) and the number of pollution energy consumption types (*DET*) are obtained from Hexun. The related regression results are shown in columns (6) to (9) of Table 9. The coefficient of *Ln_Ginvest* is significantly positive at the 5% level, indicating that the application of AI technology encourages enterprises to invest more funds in the field of environmental protection. The coefficients of *ECI*, *PT*, and *DET* are significantly negative at the 1% level, implying that the application of AI technology significantly reduces the consumption and emissions of pollution energy. Furthermore, the increase in green investment is beneficial for enterprises to engage in

Table 9
Results of mechanism analysis: the role of sustainable development strategy and green investment.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Climate	ESG	E_score	S_score	G_score	Ln_Ginvest	ECI	PT	DET
Ln_AI	0.0097***	0.0030**	0.0051***	0.0021	0.0029*	0.0439**	−0.3062***	−0.1604***	−0.1774***
	(0.0030)	(0.0013)	(0.0017)	(0.0022)	(0.0017)	(0.0223)	(0.0531)	(0.0491)	(0.0518)
Constant	−0.3272***	0.4629***	0.3964***	0.2800***	0.6089***	1.1412*	17.3057***	−4.7373***	−4.8928***
	(0.0803)	(0.0281)	(0.0345)	(0.0476)	(0.0386)	(0.6652)	(1.4239)	(0.7612)	(0.7979)
Observations	24,749	24,855	24,855	24,855	24,855	20,687	16,272	24,825	24,825
Adjusted R^2	0.7648	0.5496	0.6807	0.5733	0.4545	0.8613	0.6528	0.3816	0.3823
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

more green innovation activities, reduce the operating costs of low-carbon operations, and actively undergo energy transition (Tan et al., 2024). The decrease in the consumption and emissions of pollution energy, as the most direct influencing factors, undoubtedly promotes the energy transition. Based on the above, ***H3*** is true.

4.5. Heterogeneity analysis

Considering that the impact of technological innovation on the green transformation of enterprises may be influenced by the characteristics of the enterprises themselves, the industry they are in, and the region they operate in (Zhang et al., 2023), this section will conduct heterogeneity analysis of the impact of AI technology on corporate energy transformation from two perspectives: enterprise characteristics and industry and regional characteristics.

4.5.1. Heterogeneity analysis: characteristics of enterprise

As aforementioned, AI technology can influence energy transformation by reducing financing constraints, adjusting green development concepts, and more. However, in practice, private enterprises in China often find it more challenging to obtain bank loans and are more inclined to adjust their corporate strategies (Cull et al., 2015). This can easily lead to significant differences in the impact of AI technology on energy transformation among enterprises with different ownership structures. To verify this difference, we divide the sample into state-owned enterprises and private enterprises based on ownership and conducted regressions. The regression results are shown in Table 10, columns (1) and (2). The results indicate that *Ln_AI* has a significant promoting effect on the energy transformation of private enterprises only. This suggests that the application of AI technology helps reduce discrimination against private enterprises by external financial institutions, enabling them to obtain more loan financing and strengthen their energy transformation practices.

In addition, enterprises at different stages of their lifecycle have varying resource conditions and different risk preferences internally and externally, often holding different views on the application of new technologies and green transformation (Tan and Zhu, 2022). Considering this, the study refers to the cash flow partitioning method proposed by Dickinson (2011) and divides the enterprise lifecycle into the growth period, mature period, and decline period for sub-sample regressions. Details can be found in Table 10, columns (3) and (4). We find that the coefficient of *Ln_AI* has a significantly positive impact on energy transformation in enterprises in the growth and mature periods. This suggests that actively innovating in artificial intelligence technology during the decline period does not promote energy transformation. The possible reason is that, compared to the first two stages, enterprises in the decline period experience a decline in profitability, which hinders them from having sufficient funds to support further innovation and environmental investments, making it difficult for them to smoothly undergo energy transformation.

It is worth mentioning that customers largely determine the financial condition of enterprises. A higher concentration of customers implies a higher probability of financial and default risks for the enterprise. In such situations, enterprises often seek more cost-cutting measures to achieve stable operations. Therefore, when enterprises attempt to achieve energy transformation through applying AI, customer concentration is an essential factor. This study measures customer concentration (CC) by the proportion of sales from the enterprise's top five customers to the total sales. If this value is greater than the sample median, it belongs to the high customer concentration group, and vice versa. The grouping regression results can be found in Table 11, columns (1) and (2). It can be observed that the coefficient of *Ln_AI* is significantly positive in the High CC group. This suggests that due to the favorable impact of AI technology application on increasing economic benefits for enterprises and achieving sustainable stable operational management, enterprises with higher customer concentration are more willing to undergo energy transformation through applying AI technology.

Signal theory suggests that a company's environmental disclosure can convey a positive development signal to society, indicating that the company is actively seeking green transformation (Dhaliwal et al., 2011). Stakeholders can make more rational configurations for future investments and cash flows based on environmental behavior and performance of the corporate. This helps companies obtain more economic and innovative resources, promoting their active engagement in energy transformation. In this context, this study conducts heterogeneity analysis based on environmental disclosure. Specifically, using corporate annual reports, corporate social responsibility reports, and environmental reports as carriers of environmental information, the study obtains information on whether companies disclosed environmental concepts, environmental goals, environmental management systems, environmental education and training, special environmental actions,

Table 11
Heterogeneity analysis: customer concentration and environmental information disclosure.

Variables	(1)	(2)	(3)	(4)
	High CC	Low CC	High EID	Low EID
	Ln_ET	*Ln_ET*	*Ln_ET*	*Ln_ET*
Ln_AI	0.0536***	0.0511	0.0665***	0.0323
	(0.0182)	(0.0313)	(0.0213)	(0.0265)
Constant	−3.8741***	−2.6290***	−3.2713***	−3.0729***
	(0.5749)	(0.6972)	(0.7436)	(0.5576)
Observations	13,279	11,591	12,963	11,907
Adjusted R^2	0.8116	0.7880	0.8106	0.8022
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

Table 10
Heterogeneity analysis: enterprise ownership and lifecycle.

Variables	(1)	(2)	(3)	(4)	(5)
	SOE	Private	Growth stage	Maturity stage	Decline stage
	Ln_ET	*Ln_ET*	*Ln_ET*	*Ln_ET*	*Ln_ET*
Ln_AI	0.0299	0.0748***	0.0511**	0.0735**	0.0511
	(0.0215)	(0.0267)	(0.0221)	(0.0325)	(0.0408)
Constant	−2.5890***	−4.1408***	−3.2958***	−2.7507***	−3.6537***
	(0.8036)	(0.5571)	(0.6109)	(0.7288)	(0.8415)
Observations	9169	15,701	11,039	8869	4962
Adjusted R^2	0.7896	0.7988	0.8037	0.8005	0.7766
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

and environmental event emergency mechanisms from the CSMAR database. The sum of these aspects resulted in the environmental information disclosure (EID) of the company. Sample division is still based on the sample median, and the grouping regression results are presented in Table 11, columns (3) and (4). The results indicate that *Ln_AI* significantly promotes energy transformation only for companies in the High EID sample group. This suggests that the application of AI technology facilitates companies in releasing more positive signals for green transformation to the external environment, enabling companies with better environmental information disclosure to smoothly undergo energy transformation.

4.5.2. Heterogeneity analysis: characteristics of industries and regions

From a more macro perspective, the industrial characteristics wield prominent influence in a company's decision to achieve green transformation through technological innovation. For instance, companies in high-pollution industries may have a stronger willingness for green transformation when facing the same institutional pressures (Yang et al., 2023). Therefore, AI technology is more likely to assist companies in high-pollution industries in achieving energy transformation. To substantiate this hypothesis, the study divides corporates into high-pollution industries and non-high-pollution industries and conducted regression analysis. The regression results are presented in Table 12, columns (1) and (2). The results indeed demonstrate that AI technology is more effective in promoting energy transformation for companies in high-pollution industries.

At the same time, according to Zhang et al. (2021), the environmental governance of local governments has a significant impact on the green transformation of local businesses. Therefore, inspired by the research methodology of Chen et al. (2018), this study calculates the frequency of words related to "environmental protection" in the annual reports of provincial-level governments as a proxy variable for government environmental governance intensity (EGI). Based on the median size of environmental governance intensity, the sample data are divided into regions with high EGI and regions with low EGI. Regression analysis is then conducted to further investigate the impact of government environmental governance on the relationship between AI technology and corporate energy transformation. Please refer to Table 12, columns (3) and (4) for the regression results. The results indicate that AI technology has a more significant effect on the energy transformation of companies located in regions with low environmental governance intensity.

5. Discussion

This study aims to discuss the potential mechanism pathways of corporate energy transition against the backdrop of rapid development in AI technology. Firstly, we find that AI technology indeed significantly promotes corporate energy transition, thereby supporting ***H1***. This conclusion is partially supported by Du et al. (2023) and Wang et al. (2022a). The main reason is that the green technology innovation and energy innovation mentioned by these scholars are only one aspect of the entire process concept of energy transition. This aligns with another significant innovation point of this study, which is the reevaluation of corporate energy transition from a whole-cycle perspective (see Section 3.2.1). Consequently, this study enhances the understanding of the definition and measurement of corporate energy transition, and enriches the relevant research on the influencing factors of corporate energy transition from the perspective of AI technology development.

In addition, for the first time, this study, based on information asymmetry theory and institutional theory, finds that AI can promote corporate energy transition through the following pathways: reducing information asymmetry, alleviating financing constraints, and adjusting sustainable development concepts and practices. This constitutes an effective and positive response at the micro-level of enterprises to information asymmetry theory and institutional theory. As Cui et al. (2022) found, AI can indeed alleviate information asymmetry issues to a certain extent. Building upon the measurement of AI by Cui et al. (2022), this study supplements new evidence from patent texts, providing a more accurate demonstration of the information spillover effects of AI. Furthermore, this study suggests that AI can alleviate financing constraints and unlock significant funding potential, which partly overlaps with the viewpoint of Javaid et al. (2022), with the difference being that the latter emphasizes blockchain technology. Additionally, we discover that AI can promote energy transition by altering variables such as climate risk perception and green investment, reflecting sustainable development concepts and practices. The conclusion regarding green investment receive partial support from Lu et al. (2023). These findings not only enrich the research on the influencing factors of climate risk perception and green investment but also complement the studies of Kouloukoui et al. (2021) and Chen et al. (2023) at the forefront of the causal chain, clarifying the conclusion that AI promotes energy transition by influencing enterprise sustainable development concepts and practices.

Lastly, this study makes significant contributions by considering the heterogeneity at the levels of enterprises, industries, and regions in the literature of the impact of artificial intelligence. Drawing on the approach proposed by Zhang et al. (2023), we find heterogeneous effects of artificial intelligence on corporate energy transition at different levels. Specifically, contrary to Du et al. (2023), this study reveals that AI cannot effectively promote energy transition in state-owned enterprises. Possible reasons for this may include the current level of AI application in state-owned enterprises lagging behind that of private enterprises. Additionally, decline period enterprises may hold more negative views towards emerging technologies (Tan and Zhu, 2022), resulting in the failure of AI to drive energy transition in such enterprises, consistent with the findings of Zhai et al. (2022). As highlighted by Dhaliwal et al. (2011), signal theory suggests that enterprises with higher environmental disclosure tendencies are more inclined towards green transformation, further aiding in our understanding and interpretation of existing conclusions. This also extends the application of signal theory in corporate energy transition practices. Similar to Yang et al. (2023), enterprises in heavy pollution industries are more willing to achieve energy transition through AI. However, at the regional level, our findings contradict the viewpoint of Huang and Zou (2020), we uncover that AI is more likely to achieve energy transition in regions with weaker government environmental governance. This suggests that the adjustment of corporate sustainable development concepts may not be determined by the strength of regional environmental governance, but rather by the application of AI technology.

In summary, this study provides novel insights for government policymakers and business managers in analyzing the socioeconomic impacts of AI and the driving factors of energy transition, based on

Table 12
Heterogeneity analysis: types of industries and regions.

Variables	(1) Heavy pollution industries	(2) Non-heavy pollution industries	(3) Region with high EGI	(4) Region with low EGI
	Ln_ET	*Ln_ET*	*Ln_ET*	*Ln_ET*
Ln_AI	0.0795**	0.0330	0.0302	0.0736***
	(0.0328)	(0.0205)	(0.0234)	(0.0230)
Constant	−3.7567***	−3.0654***	−3.6004***	−3.5964***
	(0.6378)	(0.5400)	(0.5694)	(0.5602)
Observations	10,664	14,206	13,418	11,452
Adjusted R^2	0.7540	0.8103	0.8217	0.7826
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Notes: Robust standard errors clustered at the firm level and are in the parentheses; ***p < 0.01, **p < 0.05, *p < 0.1.

empirical findings and theoretical insights.

6. Conclusion and management implications

The relationship between AI and corporate energy transition, as well as the specific mechanisms, is still yet to be comprehensively explored. This study delves into the above issues based on information asymmetry theory and institutional theory, utilizing panel data from A-share listed companies on the Shanghai and Shenzhen stock exchanges in China from 2011 to 2021. We obtain several key research conclusions: Firstly, AI technology contributes to promoting corporate energy transition, and collaborative research and development innovation in AI technology can also leverage its green effects. Secondly, based on the mechanism analysis results grounded in information asymmetry theory and institutional theory, it is found that AI technology promotes corporate energy transition mainly through multiple channels, such as alleviating information asymmetry and financing constraints, adjusting corporate sustainable development concepts and practices. Additionally, heterogeneity analysis indicates that this positive effect is mainly manifested in private enterprises, non-decline period enterprises, enterprises with a higher customer concentration, enterprises with a higher environmental information disclosure, and enterprises in heavy pollution industries. Finally, in regions with lower government environmental governance intensity, the amplification of the role of AI development in promoting corporate energy transition is more pronounced.

Based on the conclusions drawn from this study, we have put forward some management and policy insights for both businesses and government, respectively, for reference.

First, we provide several management implications for enterprises managers.

(1) Corporates should increase their investments in artificial intelligence, focusing on AI systems that can enhance energy efficiency and optimize resource allocation. Particularly, for companies in the growth and maturity phases, attention should be given to upgrading AI, strengthening the integration between existing business operations and AI technology, and customized technical support is also necessary.
(2) Corporates should proactively enhance the transparency of environmental information, providing more accurate information on corporate operations and environmental risk assessments for investors and lending institutions. Additionally, after obtaining a significant amount of business loans, companies should increase the proportion of green investments to prevent a decrease in commercial loan credibility and to meet the requirements of energy transition.
(3) Corporates should enhance environmental awareness and AI knowledge through education and training for management and employees, adjusting their perception of climate risks. Proactively building a corporate culture centered around sustainable development is also crucial.

Second, we also provide several management implications for government.

(1) Government should recognize the crucial role of artificial intelligence in accelerating corporate energy transition. Policies supporting enterprise energy transition should be formulated based on the development and application characteristics of AI. This could involve optimizing credit policies to provide more favorable loan conditions for companies willing to invest in green and AI technologies. Implementing subsidy policies at the provincial and municipal levels to establish AI implementation funds for businesses can help overcome technological bottlenecks, achieve economic profitability, and facilitate energy transition. In addition, optimizing the property rights protection system is also an important policy direction.
(2) Initiating the establishment of an open platform for enterprise data, accompanied by relevant data disclosure regulations, can guide and standardize corporate operations and environmental information disclosure. This ensures the authenticity, timeliness, and accuracy of information, creating a favorable business environment while ensuring companies have sufficient funds for energy transition practices.
(3) Implementing differentiated strategic planning for regions with varying levels of environmental governance intensity and industries with different pollution levels ensures that AI can promote energy transition across a broader geographic scope. For instance, in high-pollution industries, the government can establish special funds to encourage the adoption of AI technology for environmental monitoring and pollution control. Simultaneously, promoting the upgrading of outdated equipment and optimizing clean production processes. Public-private partnership projects can attract private investment in environmental upgrades in high-pollution industries, sharing risks and rewards. For industries with lower pollution or already undergoing green transformation, policies can focus on expanding the application of AI in areas such as smart logistics, precision agriculture, and intelligent grids, enhancing efficiency while reducing environmental burdens.
(4) Facilitating the sharing of technical information and promoting business collaboration between enterprises, government, and research institutions through the creation of joint technology innovation and development centers can collectively advance the green transformation utility of AI technology in the socio-economic landscape. Regarding cooperation between enterprises, encouraging the joint construction of AI technology innovation and application systems in the supply chain can achieve cost reduction, increased efficiency, and sustainable environmental development.

The authors acknowledge the presence of limitations and potential for future in-depth exploration. Firstly, it is important to note that although this study utilizes machine learning text analysis to measure the overall situation of corporate energy transition, it has not achieved a precise characterization of energy transition in various internal processes of enterprises (such as research and development, production, sales, etc.). Future research should explore how to further characterize the energy transition in various internal processes of enterprises, allowing for a more detailed examination of the mechanisms through which the application of artificial intelligence technology influences energy transition in each process. Secondly, this study has not proven whether the AI development of small and medium-sized enterprises (SMEs) also yields green effects. In China, while listed companies may have higher capabilities to develop AI and consume more energy, numerous SMEs are significant creators of AI. Therefore, future research could incorporate different datasets to consider specific conclusions for SMEs. Lastly, at its core, AI technology represents a unique form of technological progress, implying the possibility of certain levels of technology spillover effects. In future research, we will consider incorporating supply chain data to further examine the untapped potential of AI in assisting industrial chain energy transition.

CRediT authorship contribution statement

Zhongzhu Chu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Zihan Zhang:** Writing – review & editing, Visualization, Software. **Weijie Tan:** Writing – original draft, Supervision, Formal analysis, Data curation. **Pengyu Chen:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was supported by the Major Program of the National Fund of Philosophy and Social Science of China (NO. 21&ZD123).

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